

# Minkyung Kim

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## Research Interests

Machine learning for prediction and decision-making in dynamic environments: forecasting under distribution shift, counterfactual estimation of intervention effects, and selection of interventions under resource constraints. Methods draw on graph neural networks, time series models, and large language models, with applications in public health, energy, and economics.

## Education

**Yonsei University, Graduate School of Information** Seoul, Korea  
*Integrated M.S./Ph.D. in Information Systems (expected February 2028)* Sep. 2021 – Feb. 2028  
• Advisor: Prof. Beakcheol Jang

**University of Pennsylvania, College of Arts and Sciences** Philadelphia, PA, USA  
*B.A. in Economics* May 2015

## Publications

- [1] Beyond similarity: Coverage-aware prompt selection for time series forecasting with LLMs  
D. Ji<sup>†</sup>, **M. Kim**<sup>†</sup>, D. Kim, Y. Lee, B. Kim\*, B. Jang\*  
*Findings of the Association for Computational Linguistics: EMNLP 2026*
- [2] CTRL: Control-based time series forecasting with LLM-guided residual learning  
**M. Kim**, D. Ji, Y. Lee, B. Kim, B. Jang  
*Findings of the Association for Computational Linguistics: ACL 2026*
- [3] Forecasting epidemic spread with recurrent graph gate fusion transformers  
**M. Kim**, J. H. Kim, B. Jang  
*IEEE Journal of Biomedical and Health Informatics*, 29(2), 1546–1559, 2025
- [4] Accurate total consumer price index forecasting with data augmentation, multivariate features, and sentiment analysis: A case study in Korea  
I. Seo<sup>†</sup>, **M. Kim**<sup>†</sup>, J. W. Kim, B. Jang  
*PLOS ONE*, 20(5), e0321530, 2025
- [5] COVID-19 policies and payment delays: Panel data insights from Korea’s food industry  
**M. Kim**, G. I. Kim, Y. H. Lee, B. Jang  
*International Conference on Artificial Intelligence in Information and Communication (ICAIIIC)*, 2025
- [6] An approach on improving the recommender system: Predicting movie genres based on plot summaries  
G. I. Kim, J. H. Kim, **M. Kim**, T. Kwon, B. Jang  
*International Conference on Artificial Intelligence in Information and Communication (ICAIIIC)*, 2024
- [7] Time series decomposition and hybrid RNN-based spread prediction: Focused on the international energy futures market  
**M. Kim**, S. Hwang, H. Y. Kim  
*The Korean Journal of Financial Management*, 39(4), 89–122, 2022

<sup>†</sup> Equal contribution.    \* Corresponding authors.

## Manuscripts Under Review and Preprints

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[P1] Chance-constrained selection of sequential intervention strategies from counterfactual estimates

**M. Kim**, B. Jang

*Under review. arXiv:2608.13209, 2026*

[P2] Bilevel graph structure learning, revisited: Inner-channel origins of the reported gain

**M. Kim**, B. Jang

*arXiv:2605.07577, 2026*

[P3] AutoEpiGCL: Compartmentally structured graph contrastive learning for epidemic forecasting

**M. Kim**, B. Jang

*Under review, 2026*

## Teaching

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**Incheon Academy of Science & Arts**

*Instructor, Data Analysis in Everyday Life*

Incheon, Korea

*Spring 2026*

## Academic Service

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**Conference reviewer:** NeurIPS 2026 (Evaluations and Datasets Track), AAAI 2026

**Journal reviewer:** Travel Medicine and Infectious Disease

## Professional Experience

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**Pebbles Investment**

*Assistant Manager, venture and alternative investment review*

Seoul, Korea

*Jan. 2019 – Dec. 2019*

**Hyundai Investment**

*Assistant Manager, Product Strategy Department; fund strategy simulation*

Seoul, Korea

*Jan. 2016 – Jun. 2017*

**Korea Institute of Finance**

*Research Intern, financial market policy research*

Seoul, Korea

*Jun. 2012 – Aug. 2012*

## Certifications and Skills

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**Certifications:** Financial Risk Manager (FRM)

**Programming:** Python, PyTorch, R

**Languages:** Korean (native), English (fluent)